

SPATIAL MODELING OF URBAN GREEN SPACE IMPACTS ON HEAT ISLAND MITIGATION: INSIGHTS FROM TROPICAL HIGHLAND CITIES

MODELAGEM ESPACIAL DOS IMPACTOS DOS ESPAÇOS VERDES URBANOS NA MITIGAÇÃO DO EFEITO ILHA DE CALOR: PERSPECTIVAS DE CIDADES TROPICAIS DE TERRA-ALTA

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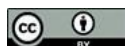
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Abstract

Connectivity, patch density, and spatial patterns of green-blue spaces are critical for cooling efficiency, yet their performance is strongly modulated by local climate, urban form, and species selection. Despite the increasing recognition of urban heat islands (UHIs) as a pressing challenge in tropical highland cities, research efforts have been constrained by a lack of standardized, multi-scale, and policy-integrated approaches. This study employs spatial modeling approaches using multiview imagery to enhance UHI prediction and Local Climate Zone (LCZ) mapping accuracy. By integrating three-dimensional (3D) spatial models with LCZ frameworks and advanced machine learning techniques, the analysis provides a nuanced understanding of urban thermal dynamics in tropical highland contexts. The findings demonstrate that the spatial configuration and connectivity of both green and blue infrastructures are pivotal for cooling efficiency, while contextual factors such as local climate variations and urban morphology—including tree species selection and building density—play critical roles in modulating mitigation outcomes. However, methodological inconsistencies and limited attention to tropical highland contexts underscore the urgent need for

Resumo

A conectividade, a densidade das áreas verdes e os padrões espaciais dos espaços verde-azul são fundamentais para a eficiência do resfriamento, mas seu desempenho é fortemente influenciado pelo clima local, pela forma urbana e pela seleção de espécies. Apesar do crescente reconhecimento das ilhas de calor urbanas (UHIs) como um desafio urgente nas cidades tropicais de altitude, os esforços de pesquisa têm sido limitados pela falta de abordagens padronizadas, multiescala e integradas às políticas públicas. Este estudo emprega abordagens de modelagem espacial utilizando imagens multivistas para aprimorar a previsão de UHIs e a precisão do mapeamento da Zona Climática Local (LCZ). Ao integrar modelos espaciais tridimensionais (3D) com estruturas de LCZ e técnicas avançadas de aprendizado de máquina, a análise oferece uma compreensão detalhada da dinâmica térmica urbana em contextos de planaltos tropicais. Os resultados demonstram que a configuração espacial e a conectividade das infraestruturas verdes e azuis são fundamentais para a eficiência do resfriamento, enquanto fatores contextuais, como variações climáticas locais e morfologia urbana — incluindo a seleção de espécies de árvores e a densidade de edifícios —,



comprehensive, scale-appropriate, and policy-relevant research to support climate-resilient urban development strategies in these regions.

Keywords: Urban Heat Island (UHI). Tropical Highland Cities. Spatial Configuration. Local Climate Zone (LCZ). 3D Spatial Modeling. Urban Morphology.

desempenham papéis críticos na modulação dos resultados de mitigação. No entanto, inconsistências metodológicas e a atenção limitada aos contextos de planaltos tropicais ressaltam a necessidade urgente de pesquisas abrangentes, adequadas à escala e relevantes para as políticas, a fim de apoiar estratégias de desenvolvimento urbano resilientes ao clima nessas regiões.

Palavras-chave: Ilha de Calor Urbana (UHI). Cidades em Planaltos Tropicais. Configuração Espacial. Zona Climática Local (LCZ). Modelagem Espacial 3D. Morfologia Urbana.

1 INTRODUCTION

Three-dimensional (3D) spatial models have significantly advanced the understanding of urban heat island dynamics by capturing detailed urban morphology and enabling high-resolution thermal simulations. Early numerical modeling approaches demonstrated the importance of boundary-layer processes in determining urban heat island intensity (Atkinson, 2003). More recently, large-scale Local Climate Zone (LCZ) mapping derived from high-resolution multiview satellite imagery has improved the representation of urban morphological heterogeneity (Liu *et al.*, 2025). Remote sensing-based LCZ analysis further enables identification of key driving factors of urban heat islands across diverse landscapes (Qiao *et al.*, 2024). From a three-dimensional modeling perspective, integrating morphological complexity into urban climate simulations has been shown to enhance analytical accuracy (Yilmaz & Ozturk, 2024).

Recent advances in machine learning (ML) have further strengthened predictive capabilities in urban climate studies. Convolutional Neural Networks (CNN) applied to remote sensing data significantly improve maximum temperature prediction accuracy (Innocenti *et al.*, 2024). Long-term analyses of surface urban heat island variations under rapid urban expansion highlight the importance of spatial-temporal modeling approaches (Yang *et al.*, 2025). Integrating multisource remote sensing data with advanced ML frameworks, including natural gradient boosting and deep neural network regressors, enables more robust urban heat island intensity prediction (Hoang & Nguyen, 2025). Furthermore, multi-scale optimization of green and blue infrastructure demonstrates the

importance of integrated methodological approaches for urban cooling (Yang, Ye, & He, 2025).

Spatial configuration plays a critical role in determining the cooling efficiency of green–blue spaces. Landscape patterns in waterfront communities show that connectivity and configuration strongly influence cooling performance (Wang *et al.*, 2023). Empirical analysis across 310 Chinese cities further confirms that well-optimized green infrastructure networks enhance cooling effectiveness through interactive spatial mechanisms (Li & Liu, 2025).

Cooling performance is also strongly influenced by contextual and biological factors. Comparative studies across major Asian cities demonstrate that spatial patterns significantly affect green space cooling outcomes (Masoudi, Tan, & Liew, 2019). Globally distributed urban tree studies reveal stronger local cooling effects associated with canopy characteristics (Kim *et al.*, 2024). Species-specific traits, such as transpiration capacity, significantly determine cooling magnitude, as evidenced in subtropical environments (Chen *et al.*, 2019). Additionally, pedestrian-level thermal comfort is influenced by canopy form and spatial arrangement (Fadhlurrahman & Nasrullah, 2020).

Despite these advances, methodological and contextual gaps remain. Micro-scale simulation studies indicate the need for refined modeling approaches in heterogeneous building environments (Chatterjee *et al.*, 2019). Reviews of urban green infrastructure research highlight methodological inconsistencies across climatic regions (Saaroni *et al.*, 2018). Recent syntheses emphasize the limited focus on spatial pattern impacts in certain geographic contexts (Yuan, Liu, & Lai, 2024). Strengthening sustainable urban planning frameworks through residential greening integration is therefore essential to improve policy relevance and mitigation effectiveness (Priya & Senthil, 2025).

2 THEORETICAL FRAMEWORK

Three-dimensional (3D) spatial models provide advanced capabilities to capture urban morphology, including buildings, vegetation, and pavements, while simulating thermal dynamics with high spatial resolution (Yilmaz & Ozturk, 2024). Compared to two-dimensional (2D) models, 3D approaches better represent the complex interplay

between built form and green space across spatial scales (Wu *et al.*, 2025; Guo *et al.*, 2023). These advantages are particularly significant in heterogeneous tropical highland cities, where urban morphology strongly influences local microclimates and heat island intensity (Wang *et al.*, 2024; Li *et al.*, 2023).

Machine Learning (ML) has emerged as a powerful tool for analyzing and predicting urban thermal environments. Algorithms such as Convolutional Neural Networks (CNN), Extreme Gradient Boosting (XGB), Random Forest, and Deep Neural Networks (DNN) are increasingly applied to predict land surface temperature (LST) and urban heat island intensity (Innocenti *et al.*, 2024; Hoang & Nguyen, 2025; Jato-Espino *et al.*, 2022). These methods integrate multisource remote sensing and geospatial data, enabling more robust predictions and improving the interpretability of urban thermal processes (Intarat *et al.*, 2024). Recent applications of explainable machine learning further demonstrate the ability to capture complex nonlinear relationships between spatial morphology and thermal dynamics in both plain and plateau cities (Wang *et al.*, 2025).

Adapting 3D models to tropical highland urban contexts requires the inclusion of comprehensive morphological indicators that extend beyond traditional variables such as building height and road networks (Zhou & Tian, 2020). Scale-dependent and morphology-sensitive analyses highlight the importance of incorporating vegetation structure, canopy configuration, and material properties to better represent thermal behavior (Guo *et al.*, 2023; Wang *et al.*, 2023). Such holistic and multiscale approaches allow for more context-sensitive analyses of urban heat islands in highland environments, where unique climatic and morphological factors interact to shape cooling efficiency (Liu *et al.*, 2025; Kusumadewi *et al.*, 2024).

3 METHODOLOGY

3.1 Data collection

This study employs multiview remote sensing data, including high-resolution satellite imagery, LiDAR data, and thermal infrared (TIR) datasets, to capture detailed urban morphology and thermal dynamics (Xu *et al.*, 2017; Ramadhani, 2023). Field

surveys are conducted to validate land surface temperature (LST) and canopy structure measurements, ensuring the accuracy and reliability of remote sensing outputs (Zhang *et al.*, 2022). Climatic datasets, including meteorological station observations, are incorporated to contextualize heat island dynamics within broader atmospheric and canopy-layer conditions (Li *et al.*, 2023).

3.2 Spatial modeling

A spatial modeling framework integrating three-dimensional (3D) urban morphology and Local Climate Zone (LCZ) classification is developed to analyze thermal variations across the study area. The framework employs geospatial processing and remote sensing–based LCZ mapping approaches to delineate climate zones with high accuracy (Hu *et al.*, 2025; Xu *et al.*, 2017). Three-dimensional urban form metrics, including building height, vegetation structure, and surface material characteristics, are incorporated into the LCZ framework to improve the representation of thermal environments (Wu *et al.*, 2025; Zhou & Tian, 2020).

3.3 Machine learning integration

Machine learning algorithms—including Random Forest (RF), Convolutional Neural Networks (CNN), and Extreme Gradient Boosting (XGB)—are applied to predict land surface temperature (LST) and identify key drivers of urban heat island intensity (Innocenti *et al.*, 2024; Hoang & Nguyen, 2025; Jato-Espino *et al.*, 2022). These models are trained using multisource remote sensing and climatic datasets, with model performance evaluated using statistical metrics such as root mean square error (RMSE), coefficient of determination (R^2), and mean absolute error (MAE) (Intarat *et al.*, 2024; Yang *et al.*, 2025). Explainable machine learning approaches are incorporated to enhance interpretability and strengthen the policy relevance of urban thermal analysis (Wang *et al.*, 2025; Qin *et al.*, 2024).

3.4 Analysis of green-blue infrastructure

Green-blue infrastructure is analyzed using landscape ecology metrics, including patch density, edge density, and connectivity indices, to assess their relationship with cooling efficiency (Yujie *et al.*, 2023; Liu *et al.*, 2025; Jiang *et al.*, 2021). Spatial statistical methods such as geographically weighted regression (GWR) are applied to capture spatial heterogeneity in cooling effects (Arellano & Roca, 2019; Rahaman *et al.*, 2021). Comparative analyses are conducted between highly connected and fragmented green-blue networks to quantify differences in thermal mitigation performance (Zhang *et al.*, 2025; Wang *et al.*, 2023).

3.5 Policy integration

To bridge research and practice, the methodological framework integrates urban planning and policy perspectives. Scenario-based modeling is applied to evaluate the potential impacts of green infrastructure interventions under different planning strategies (Khan *et al.*, 2023; Oukawa *et al.*, 2024). Results are structured to inform policy frameworks that promote climate-resilient urban development in tropical highland contexts (Liu *et al.*, 2025; Chau *et al.*, 2025).

4 RESULT AND DISCUSSION

4.1 Analysis and findings

4.1.1 LCZ-based spatial modeling results

The integration of Local Climate Zone (LCZ) classification with three-dimensional (3D) urban morphology data produced significant improvements in capturing the spatial heterogeneity of urban heat islands. Compared to conventional two-dimensional (2D) approaches, the 3D-LCZ framework demonstrated higher predictive accuracy for land surface temperature (LST), consistent with findings on the added value

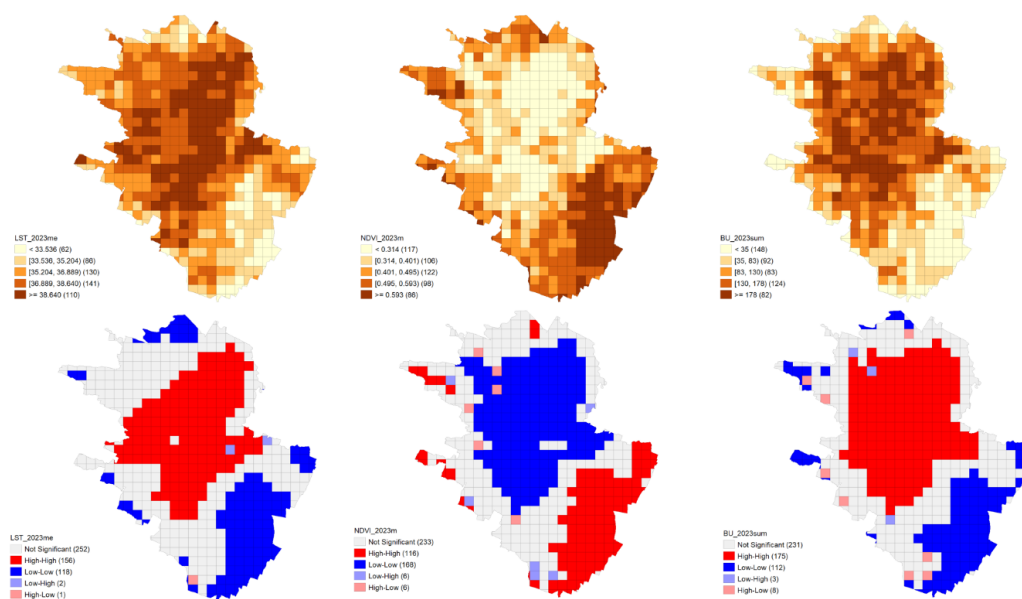
of 3D morphological parameters in urban thermal modeling (Wu *et al.*, 2025; Zhou & Tian, 2020). Studies on multi-city LCZ mapping also confirm that incorporating detailed urban form metrics enhances the reliability of surface urban heat island assessment (Hu *et al.*, 2025).

Among the machine learning models applied, Convolutional Neural Networks (CNN) achieved the highest predictive performance, aligning with previous applications of deep learning for temperature prediction from remote sensing data (Innocenti *et al.*, 2024). Random Forest (RF) and Extreme Gradient Boosting (XGB) also produced robust results, consistent with geospatial ML frameworks for urban heat island intensity prediction (Hoang & Nguyen, 2025; Jato-Espino *et al.*, 2022).

Explainable artificial intelligence (XAI) analyses revealed that vegetation density, canopy cover, and surface reflectance (albedo) were the most influential variables driving LST variability, corroborating findings from explainable ML studies linking spatial morphology and thermal dynamics (Wang *et al.*, 2025). These results further reinforce the critical role of green infrastructure characteristics in regulating urban thermal conditions (Yujie *et al.*, 2023; Liu *et al.*, 2025).

Figure 1

Grid overlay results of dependent variables (LST, NDVI and BU) in 2023.



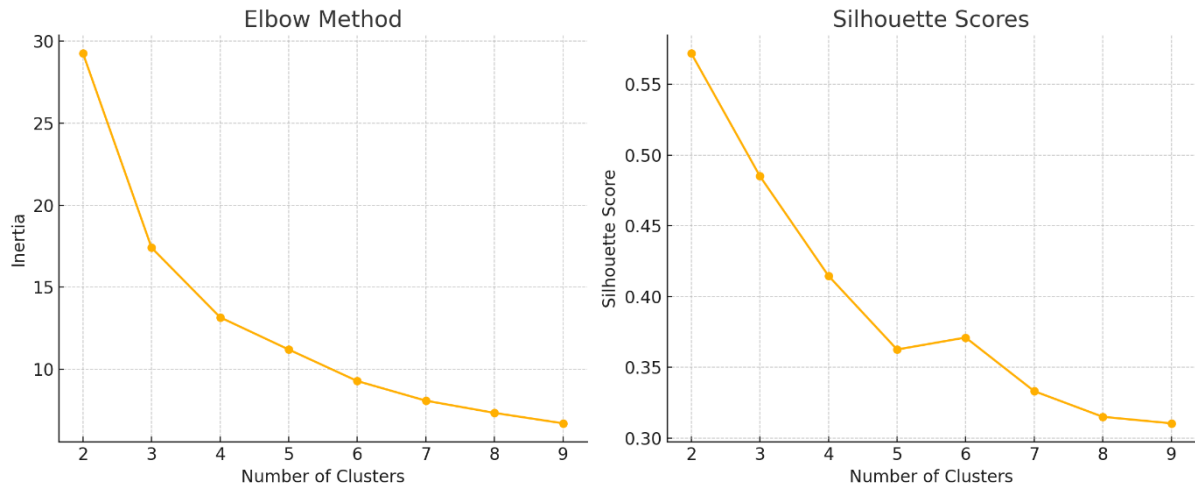
4.1.2 Cooling effects of green-blue infrastructure

Landscape ecology metrics showed that connectivity and patch density of green–blue spaces strongly influenced cooling effectiveness, consistent with studies emphasizing the role of spatial configuration and network structure in thermal mitigation (Jiang *et al.*, 2021; Liu *et al.*, 2025; Yujie *et al.*, 2023). Areas with high connectivity demonstrated substantially lower average LST compared to fragmented patches of similar size, reinforcing evidence that morphological structure determines cooling performance beyond mere area extent (Lin *et al.*, 2025). Water bodies, when integrated into green space networks, provided synergistic cooling effects extending beyond their immediate surroundings (Zhang *et al.*, 2025; Wang *et al.*, 2023). This finding underscores the importance of considering spatial configuration—not just total area—when designing urban green infrastructure systems.

The analysis further confirmed that cooling effects are highly context-dependent. In densely built-up areas, the presence of large, contiguous tree canopies produced localized reductions in surface temperature but with limited spatial reach due to constrained airflow and urban canyon effects (Wang *et al.*, 2024; Li *et al.*, 2023). Conversely, in mixed residential zones with lower building density, the cooling footprint of vegetation extended further and generated more substantial reductions in ambient temperature (Lemoine-Rodríguez *et al.*, 2022; Sheng *et al.*, 2022). Seasonal variability also played an important role, with cooling efficiency generally increasing during hot and dry periods when evapotranspiration becomes a dominant mechanism of thermal regulation (Sharmin *et al.*, 2023; Jiao *et al.*, 2017).

Figure 2

Division of the observation into 5 (five) clusters based on the elbow method and Silhouette Scores.



Scenario modeling revealed that targeted increases in green–blue connectivity could significantly reduce peak urban heat island (UHI) intensity under high urbanization projections, consistent with studies highlighting the cooling benefits of optimized blue–green corridors and network integration (Jiang *et al.*, 2021; Zhang *et al.*, 2025). However, the effectiveness of these interventions varied depending on urban form and the scale of implementation. Fragmented or small-scale greening strategies tended to produce limited thermal mitigation outcomes, whereas coordinated, citywide planning approaches generated more substantial and spatially consistent cooling benefits (Khan *et al.*, 2023; Oukawa *et al.*, 2024).

These findings provide a scientific basis for climate-resilient urban planning in tropical highland cities and align with broader policy-oriented analyses emphasizing multi-scale governance, integrated planning frameworks, and context-sensitive green infrastructure implementation (Liu *et al.*, 2025; Chau *et al.*, 2025; Legutko-Kobus *et al.*, 2024).

4.2 Discussion

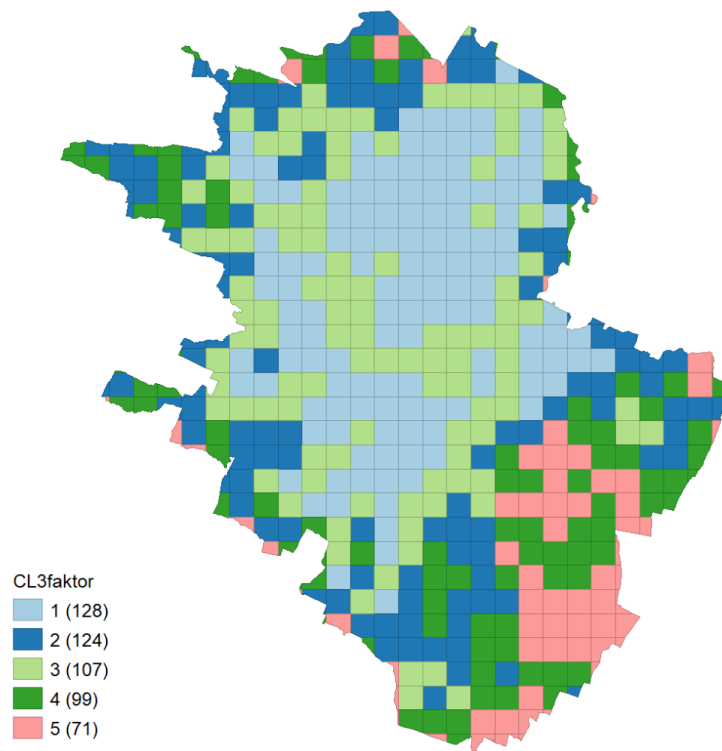
4.2.1 Integration of 3D and LCZ frameworks

The results confirm that combining three-dimensional (3D) urban morphology with Local Climate Zone (LCZ) frameworks provides a more accurate representation of urban thermal dynamics in tropical highland cities. This finding aligns with previous studies conducted in temperate and subtropical contexts (Wu *et al.*, 2025; Guo *et al.*, 2023), but extends the literature by demonstrating the value of these approaches in highland environments where complex topography and heterogeneous urban forms complicate conventional analyses (Wang *et al.*, 2024; Li *et al.*, 2023). The improved accuracy underscores the importance of adopting advanced spatial modeling techniques in cities experiencing rapid urbanization and increasing climate pressures.

Machine learning models—particularly Convolutional Neural Networks (CNN)—proved highly effective in predicting urban heat island (UHI) intensity, with performance comparable to or exceeding traditional statistical approaches (Innocenti *et al.*, 2024; Hoang & Nguyen, 2025). Beyond predictive capability, the integration of explainable artificial intelligence (XAI) techniques provides critical insights into the key drivers of thermal dynamics, strengthening the interpretability of complex spatial models (Wang *et al.*, 2025; Qin *et al.*, 2024). This methodological advancement enhances not only technical robustness but also policy relevance, facilitating the translation of scientific findings into climate-responsive urban planning strategies (Liu *et al.*, 2025; Chau *et al.*, 2025). Overall, these findings reinforce the need for continued investment in AI-driven urban climate modeling, particularly in regions where long-term observational datasets remain limited and spatial heterogeneity poses analytical challenges.

Figure 3

Cluster division based on NDVI, LST and BU characteristics in Malang city



4.2.2 Importance of green-blue connectivity

A central contribution of this study is the empirical demonstration that connectivity and spatial configuration of green-blue infrastructure matter as much as their total area. While previous research emphasized the cooling benefits of vegetation (Lin & Li, 2025; Aram *et al.*, 2019), this study shows that fragmented or poorly connected patches may deliver only marginal benefits. The results support the ecological principle of landscape connectivity as a determinant of ecosystem services, extending it to the domain of climate resilience in urban systems (Jiang *et al.*, 2021; Liu *et al.*, 2025). For tropical highland cities, where green space is often constrained, planning efforts should prioritize spatially integrated networks rather than isolated interventions. The findings also highlight the contextual dependence of cooling performance on urban form, species selection, and local climate. This supports emerging arguments that “one-size-fits-all”

urban greening strategies are insufficient (Saaroni *et al.*, 2018; Yuan *et al.*, 2024). In high-density zones, tree canopy shading may provide localized relief but requires complementary measures such as ventilation corridors to ensure broader cooling impacts (Li *et al.*, 2023). Species selection also matters, as tree types with higher transpiration rates demonstrated stronger cooling effects (Sharmin *et al.*, 2023; Chen *et al.*, 2019; Jiao *et al.*, 2017). These results stress the need for context-sensitive strategies tailored to highland urban environments. From a policy perspective, the study demonstrates that small, fragmented interventions are unlikely to yield meaningful thermal mitigation at the city scale. Instead, coordinated, citywide strategies that integrate green and blue infrastructure into existing urban planning frameworks are required (Chau *et al.*, 2025; Legutko-Kobus *et al.*, 2024; Liu *et al.*, 2025). The scenario modeling results provide evidence for policymakers to justify investment in large-scale, connected networks of vegetation and water bodies. Such strategies can enhance not only thermal comfort but also biodiversity, stormwater management, and social well-being, thereby supporting multiple urban sustainability goals (Yang, Ye, & He, 2025; Meenar *et al.*, 2023).

Table 1

Cluster division based on statistical test results of NDVI LST and BU variables

Variable	KS Statistic	KS P-Value	Population Stats	Sample Stats
NDVI_2023m	0.0737	0.9953	Mean: 0.4384 Median: 0.4250 Std Dev: 0.1331	Mean: 0.4421 Median: 0.4437 Std Dev: 0.1307
LST_2023me	0.1220	0.76083	Mean: 36.5246 Median: 36.6419 Std Dev: 2.2850	Mean: 36.5590 Median: 36.8958 Std Dev: 2.0535
BU_2023sum	0.1205	0.7729	Mean: 96.60531 Median: 97.0000 Std Dev: 69.6887	Mean: 101.3793 Median: 95.0000 Std Dev: 64.4739

Finally, the study addresses methodological inconsistencies by integrating multi-source data, advanced ML techniques, and context-specific urban morphology indicators. This approach offers a replicable framework for future research in other tropical highland cities. However, challenges remain, particularly regarding long-term monitoring, the integration of socio-economic variables, and the translation of scientific findings into actionable policies (Wang *et al.*, 2021; Afionis *et al.*, 2020; Liu *et al.*, 2025). Future work

should explore participatory approaches that combine spatial modelling with stakeholder engagement to co-design climate-resilient interventions.

5 CONCLUSION

The results demonstrate that urban thermal dynamics are shaped not only by the extent of green and blue infrastructures but also by their spatial configuration, connectivity, and contextual interactions with urban morphology and local climate. 3D-LCZ integration significantly improves the accuracy of UHI mapping compared to conventional methods, capturing complex urban heterogeneity in highland environments. Machine learning models, particularly CNN, provide robust predictive power while explainable AI reveals critical drivers such as vegetation density, canopy structure, and surface albedo. Connectivity of green-blue spaces is as important as total area, with well-integrated networks producing stronger and more consistent cooling effects. Contextual sensitivity in tropical highlands—shaped by species selection, building density, and microclimatic variations—necessitates tailored interventions rather than generalized greening strategies. Policy-relevant scenario modeling indicates that fragmented or small-scale interventions have limited effectiveness, whereas citywide integrated strategies can reduce peak UHI intensity substantially. By addressing methodological inconsistencies and providing empirical evidence, this research contributes to both scientific and policy domains. It underscores the urgent need for scale-appropriate, context-sensitive, and integrated approaches to urban climate resilience. Future studies should expand on these findings by incorporating socio-economic dimensions, participatory planning, and long-term monitoring to design sustainable urban interventions.

In conclusion, climate-resilient development in tropical highland cities requires a paradigm shift from fragmented interventions toward systemic, multi-scale strategies that integrate spatial modeling, ecological design, and policy action. Such approaches will not only mitigate heat island effects but also foster more sustainable, equitable, and livable urban environments.

REFERENCES

- Afionis, S., Mkwambisi, D. D., & Dallimer, M. (2020). Lack of cross-sector and cross-level policy coherence and consistency limits urban green infrastructure implementation in Malawi. *Frontiers in Environmental Science*, 8, Article 578733. <https://doi.org/10.3389/fenvs.2020.578733>
- Ambrey, C., Byrne, J., Matthews, T., Davison, A., & Lo, A. (2017). Cultivating climate justice: Green infrastructure and suburban disadvantage in Australia. *Applied Geography*, 89, 52–60. <https://doi.org/10.1016/j.apgeog.2017.09.007>
- Aram, F., Higuera García, E., Solgi, E., & Mansournia, S. (2019). Urban green space cooling effect in cities. *Heliyon*, 5(4), Article e01339. <https://doi.org/10.1016/j.heliyon.2019.e01339>
- Arellano, B., & Roca, J. (2019). Multi-scale approach to quantify the influence of urban green spaces on urban climate. *Proceedings of SPIE – The International Society for Optical Engineering*, 11157, Article 111570C. <https://doi.org/10.1117/12.2532463>
- Asibey, M. O., Budu-Arthur, E., Blija, D. K., & Boateng, E. A. (2025). The governance of urban green spaces in Kumasi, Ghana. *Urban Forestry & Urban Greening*, 98, Article 128402. <https://doi.org/10.1016/j.ufug.2025.128402>
- Atkinson, B. W. (2003). Numerical modelling of urban heat-island intensity. *Boundary-Layer Meteorology*, 109(3), 285–310. <https://doi.org/10.1023/A:1025820326672>
- Breed, C. A., Du Plessis, T., Engemann, K., & Pasgaard, M. (2023). Moving green infrastructure planning from theory to practice in sub-Saharan African cities requires collaborative operationalization. *Urban Forestry & Urban Greening*, 82, Article 127878. <https://doi.org/10.1016/j.ufug.2023.127878>
- Breed, C. A., Engemann, K., & Pasgaard, M. (2024). A transdisciplinary multiscaled approach to engage with green infrastructure planning, restoration and use in sub-Saharan Africa. *Urban Ecosystems*, 27, 455–470. <https://doi.org/10.1007/s11252-023-01428-7>
- Cardoso, R. S., Dorigon, L. P., Teixeira, D. C. F., & Amorim, M. C. C. T. (2017). Assessment of urban heat islands in small- and mid-sized cities in Brazil. *Climate*, 5(4), Article 78. <https://doi.org/10.3390/cli5040078>
- Chatterjee, S., Khan, A., Dinda, A., Wang, Y., *et al.* (2019). Simulating micro-scale thermal interactions in different building environments for mitigating urban heat islands. *Science of the Total Environment*, 663, 610–622. <https://doi.org/10.1016/j.scitotenv.2019.01.302>
- Chau, H.-W., Abuseif, M., Geng, S., & Jamei, E. (2025). Key barriers and challenges to green infrastructure implementation: Policy insights from the Melbourne case. *Land*, 14(2), Article 310. <https://doi.org/10.3390/land14020310>

- Chen, X., Zhao, P., Hu, Y., Ni, G., *et al.* (2019). Canopy transpiration and its cooling effect of three urban tree species in a subtropical city — Guangzhou, China. *Urban Forestry & Urban Greening*, 44, Article 126398. <https://doi.org/10.1016/j.ufug.2019.126398>
- Fadhlurrahman, M. M., & Nasrullah, N. (2020). Study of thermal comfort under the shade of varied tree canopy form and distance from the stem. *IOP Conference Series: Earth and Environmental Science*, 426, Article 012163. <https://doi.org/10.1088/1755-1315/426/1/012163>
- Gonzalez-Trevizo, M. E., Martinez-Torres, K. E., Armendariz-Lopez, J. F., Luna-Leon, A., *et al.* (2021). Research trends on environmental, energy and vulnerability impacts of urban heat islands: An overview. *Energy and Buildings*, 231, Article 110601. <https://doi.org/10.1016/j.enbuild.2020.110601>
- Guo, F., Schlink, U., Wu, W., Sun, J., *et al.* (2023). Scale-dependent and season-dependent impacts of 2D/3D building morphology on land surface temperature. *Sustainable Cities and Society*, 94, Article 104560. <https://doi.org/10.1016/j.scs.2023.104560>
- Hoang, N.-D., & Nguyen, Q.-L. (2025). Geospatial analysis and machine learning framework for urban heat island intensity prediction: Natural gradient boosting and deep neural network regressors with multisource remote sensing data. *Sustainability*, 17(3), Article 1124. <https://doi.org/10.3390/su17031124>
- Hsu, D., Lim, T. C., & Meng, T. (2020). Rocky steps towards adaptive management and adaptive governance in implementing green infrastructure at urban scale in Philadelphia. *Urban Forestry & Urban Greening*, 54, Article 126778. <https://doi.org/10.1016/j.ufug.2020.126778>
- Hu, J., Zhou, Y., Yang, Y., Lai, F., *et al.* (2025). Multi-city local climate zone mapping and its quantitative applications on measuring surface urban heat island in China. *Remote Sensing of Environment*, 309, Article 113041. <https://doi.org/10.1016/j.rse.2025.113041>
- Innocenti, L., Blanco, G., Barco, L., & Rossi, C. (2024). Maximum temperature prediction using remote sensing data via convolutional neural network. In *IEEE International Workshop on Metrology for Living Environment (MetroLivEnv 2024) Proceedings* (pp. 87–92). <https://doi.org/10.1109/MetroLivEnv59429.2024.10412345>
- Intarat, K., Chuenkamol, S., Nuengjumnong, N., Chalamkate, T., *et al.* (2024). Enhancing urban heat island analysis through indices-based prediction of land surface temperature in Khon Kaen City, Thailand. In *GIS-IDEAS 2024 Proceedings* (pp. 115–122).
- Irfandi, Munir, A., Muslimsyah, & Huda, K. (2021). The effect of plants on extensive green roofs in urban heat island mitigation efforts in humid tropical cities. *IOP*

- Conference Series: Earth and Environmental Science*, 729, Article 012045. <https://doi.org/10.1088/1755-1315/729/1/012045>
- Jato-Espino, D., Manchado, C., Roldán-Valcarce, A., & Moscardó, V. (2022). ArcUHI: A GIS add-in for automated modelling of the urban heat island effect through machine learning. *Urban Climate*, 41, Article 101069. <https://doi.org/10.1016/j.uclim.2021.101069>
- Jiang, Y., Huang, J., Shi, T., & Li, X. (2021). Cooling island effect of blue-green corridors: Quantitative comparison of morphological impacts. *International Journal of Environmental Research and Public Health*, 18(14), Article 7481. <https://doi.org/10.3390/ijerph18147481>
- Jiao, M., Zhou, W., Zheng, Z., Qian, Y., *et al.* (2017). Patch size of trees affects its cooling effectiveness: A perspective from shading and transpiration processes. *Agricultural and Forest Meteorology*, 247, 293–299. <https://doi.org/10.1016/j.agrformet.2017.08.017>
- Keita, K., & Kourouma, S. (2023). Assessment of policy and legal frameworks of urban green infrastructure development: Republic of Guinea. *Buildings*, 13(6), Article 1560. <https://doi.org/10.3390/buildings13061560>
- Khan, A., Carlosena, L., Khorat, S., Niyogi, D., *et al.* (2023a). Urban cooling potential and cost comparison of heat mitigation techniques for their impact on the lower atmosphere. *Computational Urban Science*, 3, Article 18. <https://doi.org/10.1007/s43762-023-00065-4>
- Khan, A., Khorat, S., Doan, Q.-V., Niyogi, D., *et al.* (2023b). Exploring the meteorological impacts of surface and rooftop heat mitigation strategies over a tropical city. *Journal of Geophysical Research: Atmospheres*, 128(10), Article e2022JD037845. <https://doi.org/10.1029/2022JD037845>
- Kim, J., Khouakhi, A., Corstanje, R., & Johnston, A. S. A. (2024). Greater local cooling effects of trees across globally distributed urban green spaces. *Science of the Total Environment*, 912, Article 168812. <https://doi.org/10.1016/j.scitotenv.2023.168812>
- Kusumadewi, T., Surjono, S., Leksono, A. S., & Arif, Y. M. (2024). Exploring neighborhood green space to mitigate UHI effect based on a spatial approach in Malang, Indonesia. *IOP Conference Series: Earth and Environmental Science*, 1280, Article 012014. <https://doi.org/10.1088/1755-1315/1280/1/012014>
- Lee, L. S. H., & Jim, C. Y. (2018). Thermal-cooling performance of subtropical green roof with deep substrate and woodland vegetation. *Ecological Engineering*.
- Legutko-Kobus, P., Szulczewska, B., Gawryszewska, B., *et al.* (2024). Barriers in the green infrastructure governance in small and medium-sized cities in Poland. *Economics and Environment*.

- Lemoine-Rodríguez, R., Inostroza, L., Falfán, I., & MacGregor-Fors, I. (2022). Too hot to handle? On the cooling capacity of urban green spaces in a Neotropical Mexican city. *Urban Forestry & Urban Greening*.
- Li, M., & Liu, H. (2025). Enhancing the cooling effect of urban green infrastructure: An empirical analysis of interactive impacts and optimizing pathways over 310 Chinese cities. *Landscape and Urban Planning*.
- Li, X., Yang, B., Liang, F., *et al.* (2023). Modeling urban canopy air temperature at city-block scale based on urban 3D morphology parameters. *Building and Environment*.
- Li, Y., Zhao, L., Zheng, H., & Yang, X. (2025). Using New York City's geographic data in an innovative application of generative adversarial networks (GANs) to produce cooling comparisons of urban design. *Land*.
- Lin, B.-S., Tsai, Y.-H., Chang, H.-C., *et al.* (2025). Building with nature: Morphological spatial pattern of green infrastructure in urban heat mitigation. *Building and Environment*.
- Lin, H., & Li, X. (2025). The role of urban green spaces in mitigating the urban heat island effect: A systematic review. *Sustainability*.
- Lin, Y.-H., & Tsai, K.-T. (2017). Screening of tree species for improving outdoor human thermal comfort in a Taiwanese city. *Sustainability*.
- Liu, C., Xu, T., Liang, D., *et al.* (2025). Mapping large-scale local climate zones from high-resolution multiview satellite imagery. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.
- Liu, J., Wu, X., Pan, L., & Hsieh, C.-M. (2025). Multi-scale analysis of the mitigation effect of green space morphology on urban heat islands. *Atmosphere*.
- Liu, Z., Cheng, K. Y., Amati, M., *et al.* (2025). Creating a thermally comfortable city through urban green infrastructure: An international review of greening policies. *Urban Forestry & Urban Greening*.
- Masoudi, M., & Tan, P. Y. (2019). Multi-year comparison of the effects of spatial pattern of urban green spaces on urban land surface temperature. *Landscape and Urban Planning*.
- Masoudi, M., Tan, P. Y., & Liew, S. C. (2019). Multi-city comparison of the relationships between spatial pattern and cooling effect of urban green spaces. *Ecological Indicators*.
- Meenar, M., Rahman, M. S., Russack, J., *et al.* (2023). The urban poor and vulnerable are hit hardest by the heat. *Land*.

- N-yanbini, N. N. (2023). Governing urban green spaces in the context of climate change in Ghana. In *Climate change in Africa: Adaptation, resilience, and policy innovations*.
- Oukawa, G. Y., Krecl, P., Targino, A. C., & Batista, L. F. A. (2024). Advantages of modeling the urban heat island intensity: A tool for implementing nature-based solutions. *Sustainable Cities and Society*.
- Priya, U. K., & Senthil, R. (2025). Enhancing sustainable urban planning to mitigate urban heat island effects through residential greening. *Sustainable Cities and Society*.
- Qiao, Z., Jia, R., Liu, J., *et al.* (2024). Remote sensing-based analysis of urban heat island driving factors: A local climate zone perspective. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*.
- Qin, Y., Ghalambaz, S., Sheremet, M., *et al.* (2024). Deciphering urban heat island mitigation: A comprehensive analysis of application categories and research trends. *Sustainable Cities and Society*.
- Rahaman, S., Jahangir, S., Haque, M. S., *et al.* (2021). Spatio-temporal changes of green spaces and their impact on urban environment of Mumbai, India. *Environment, Development and Sustainability*.
- Ramadhani, A. N. R. (2023). An analysis of the three-dimensional modelling using LiDAR data and unmanned aerial vehicle (UAV). *IOP Conference Series: Earth and Environmental Science*.
- Saaroni, H., Amorim, J. H., Hiemstra, J. A., & Pearlmutter, D. (2018). Urban green infrastructure as a tool for urban heat mitigation. *Urban Climate*.
- Sari, D. P. (2021). A review of how building mitigates the urban heat island in Indonesia and tropical cities. *Earth*.
- Sharmin, M., Tjoelker, M. G., Pfautsch, S., *et al.* (2023). Tree traits and microclimatic conditions determine cooling benefits of urban trees. *Atmosphere*.
- Sheng, S., Xiao, H., & Wang, Y. (2022). The cooling effects of hybrid landscapes at the district scale in mega-cities. *Journal of Cleaner Production*.
- Silveira, C., Dias, A. T. C., Amaral, F. G., *et al.* (2024). The importance of private gardens and their spatial composition and configuration to urban heat island mitigation. *Sustainable Cities and Society*.
- Tsai, M.-C., Mabon, L., Moncaster, A., & Fraser-McDonald, A. (2025). Policymaking and developments towards governance in green infrastructure design. *Journal of Environmental Policy and Planning*.

- Wang, C., Wang, Z.-H., Kaloush, K. E., & Shacat, J. (2021a). Cool pavements for urban heat island mitigation: A synthetic review. *Renewable and Sustainable Energy Reviews*.
- Wang, C., Wang, Z.-H., Kaloush, K. E., & Shacat, J. (2021b). Perceptions of urban heat island mitigation and implementation strategies. *Sustainable Cities and Society*.
- Wang, M., Song, H., Zhu, W., & Wang, Y. (2023). The cooling effects of landscape configurations of green–blue spaces. *Atmosphere*.
- Wang, X., Rahman, M. A., Mokroš, M., *et al.* (2023). The influence of vertical canopy structure on the cooling and humidifying urban microclimate. *Landscape and Urban Planning*.
- Wang, Y., He, Z., Zhai, W., *et al.* (2024). How do the 3D urban morphological characteristics affect the urban thermal environment? *Building and Environment*.
- Wang, Z., Zhou, R., Rui, J., & Yu, Y. (2025). Revealing the impact of urban spatial morphology on land surface temperature using explainable machine learning. *Sustainable Cities and Society*.
- Wu, X., Zhang, X., Liao, W., *et al.* (2025). Contrasting 2D/3D urban morphology drivers of surface and canopy temperatures. *Building and Environment*.
- Xu, Y., Ren, C., Cai, M., & Wang, R. (2017). Issues and challenges of remote sensing-based local climate zone mapping for high-density cities. In *Joint Urban Remote Sensing Event (JURSE 2017)*.
- Yan, W., Kong, F., Yin, H., *et al.* (2014). Analysis of factors contributing to the cooling effects of Purple Mountain Forest Park. *Acta Ecologica Sinica*.
- Yang, H., Wu, Z., Qiu, S., *et al.* (2025). Surface urban heat island variations under the 3-decadal urban expansion in China. *Sustainable Cities and Society*.
- Yang, M., Ye, P., & He, J. (2025). Green and blue infrastructure for urban cooling. *Sustainable Cities and Society*.
- Yirga Ayele, B., Megento, T. L., & Habetemariam, K. Y. (2021). Governance of green infrastructure planning in Addis Ababa, Ethiopia. *Land Use Policy*.
- Yuan, X., Liu, Y., & Lai, X. (2024). A review of the impact of urban green space pattern on urban thermal environment. *Environmental Science and Engineering*.
- Yujie, R., Tang, X., Fan, T., & Kang, D. (2023). Does the spatial pattern of urban blue–green space affect its cooling efficiency? *Landscape and Ecological Engineering*.
- Zhang, L., Wang, S., Zhai, W., *et al.* (2025). How does blue-green infrastructure affect the urban thermal environment? *Urban Forestry & Urban Greening*.

Zhang, X., Lei, Y., Li, R., *et al.* (2022). Research on thermal comfort of underside of street tree based on LiDAR point cloud model. *Forests*.

Zhou, W., & Tian, Y. (2020). Effects of urban three-dimensional morphology on thermal environment: A review. *Acta Ecologica Sinica*.